

ATENEO DE ZAMBOANGA UNIVERSITY



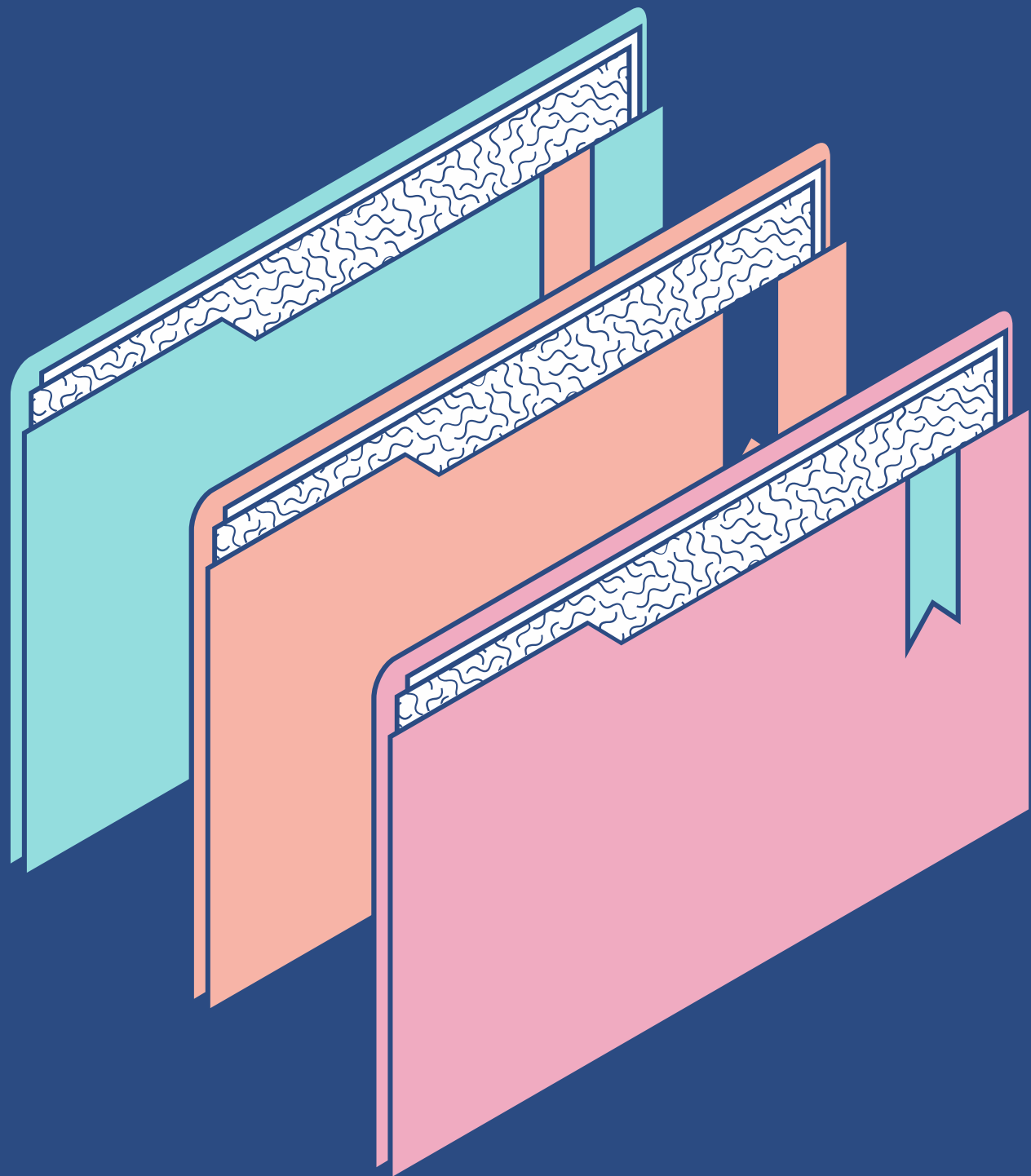
Digital vs. Analog Signals

ALEEKHAZER J. JAUSAN

CIT.016 | DATA COMMUNICATION



ATENEO DE ZAMBOANGA UNIVERSITY
CIT.016 | DATA COMMUNICATION



Digital vs. Analog Signals



But first...

What do digital and analog mean?

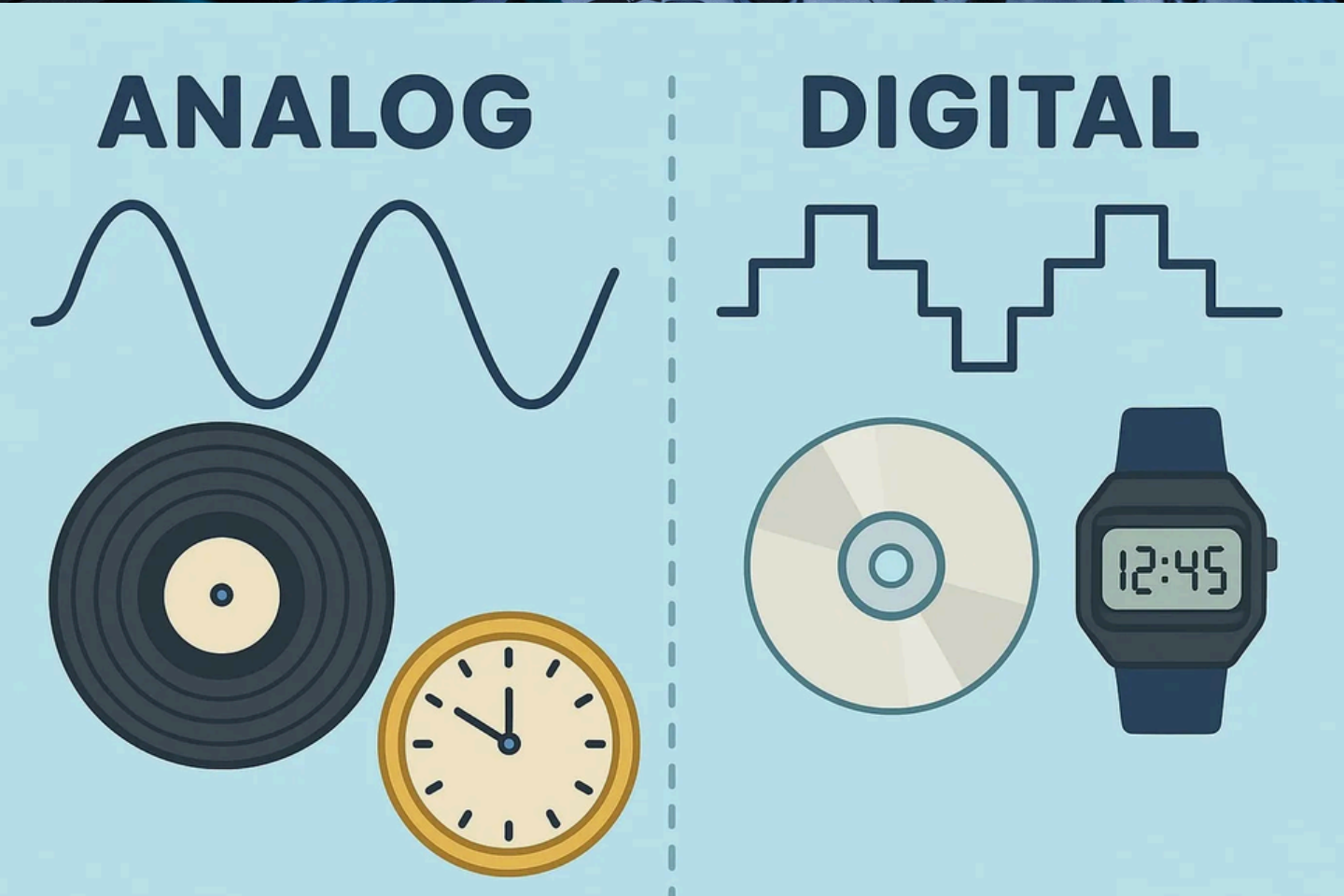


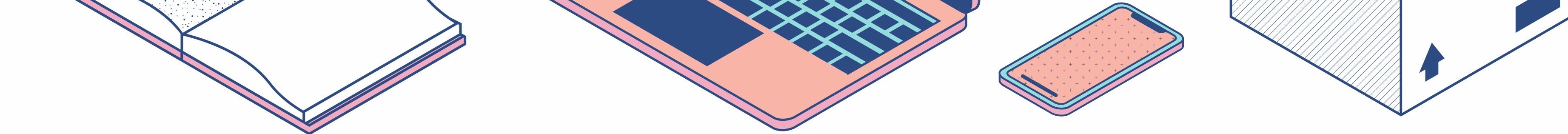
Digital and analog...

are two ways to represent, store, and transmit information.

Analog systems use continuous, smooth physical quantities—like waves or voltages—to mirror real-world changes in sound, light, or temperature.

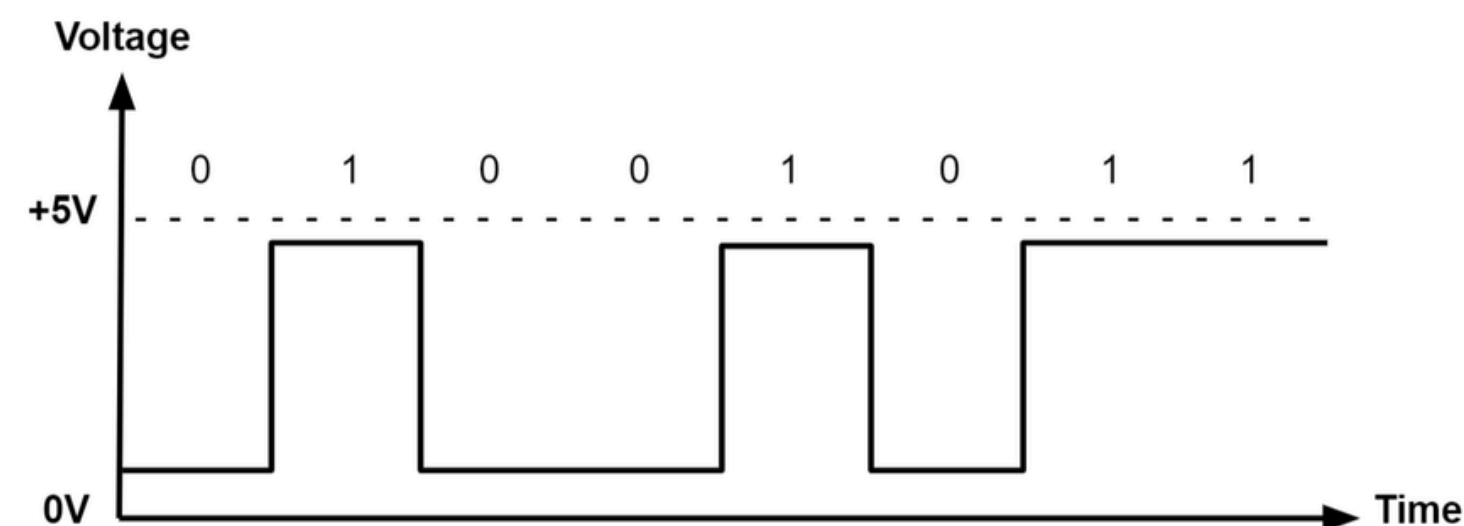
Digital systems convert this information into distinct, discontinuous steps or numbers, typically using binary code (0s and 1s).





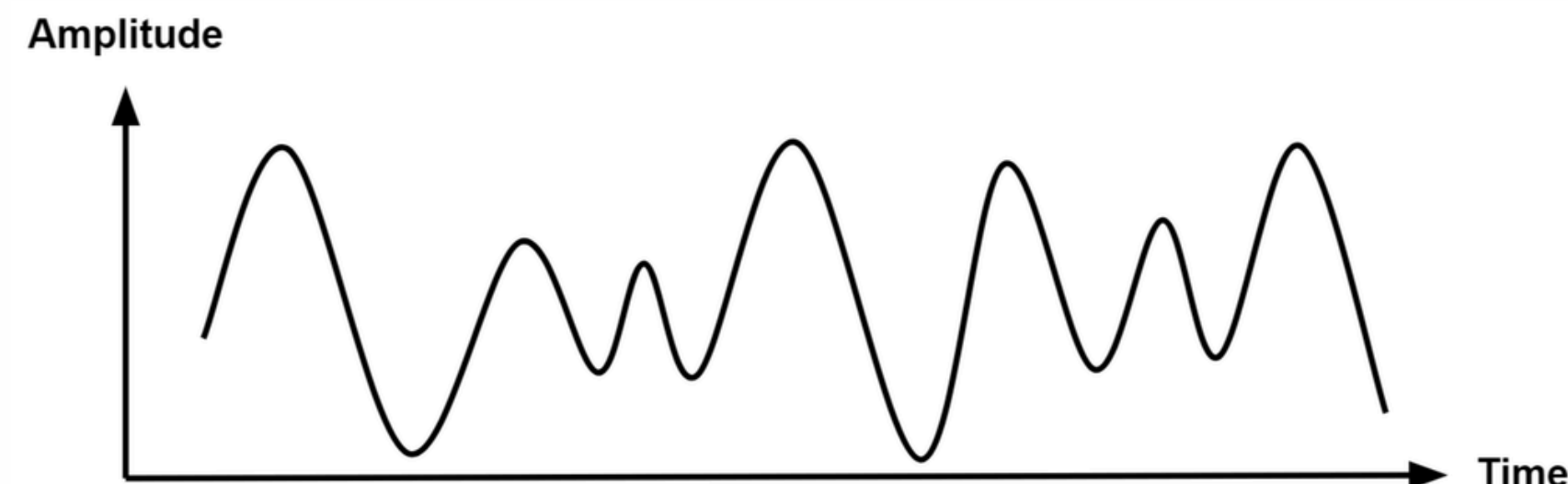
Digital Signals

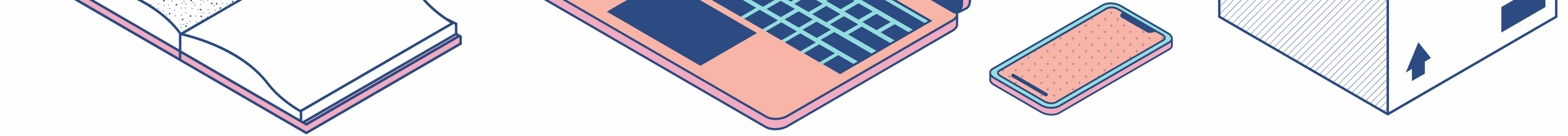
- sequence of **discrete** values, taking on one of a finite set of possible values at any given time
- **USES:** modern electronics such as telecommunications, computers and digital media



Analog Signals

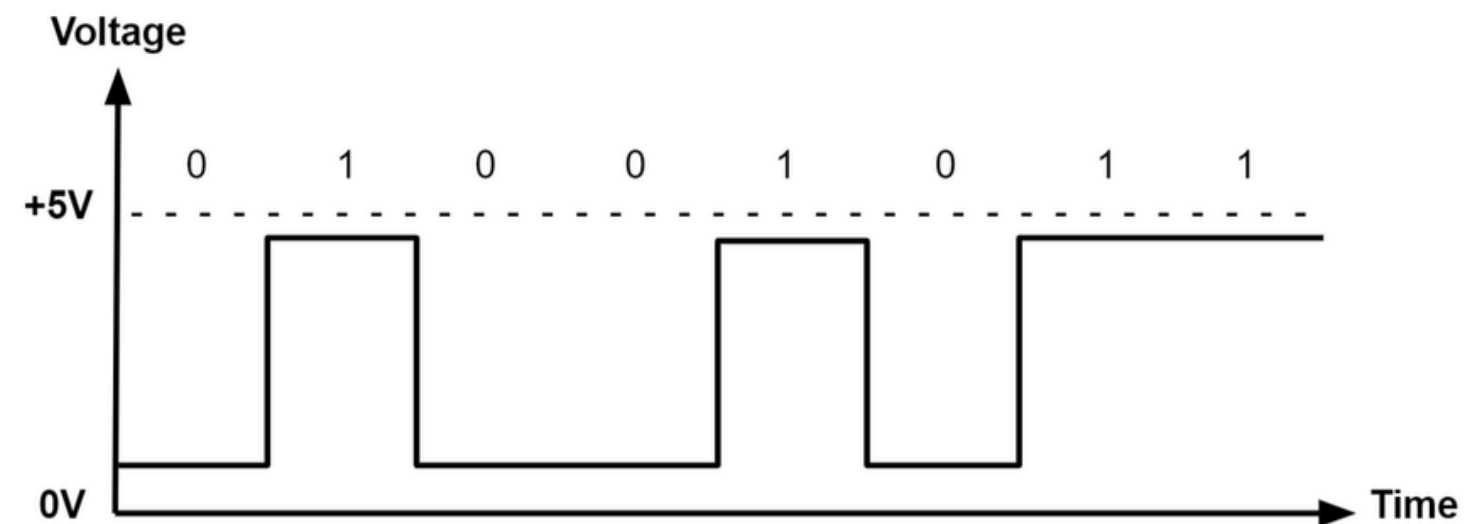
- **continuous**, time-varying signals that can take on an infinite number of values within a certain range
- **USES:** traditional technologies such as radio broadcasting, and older television systems





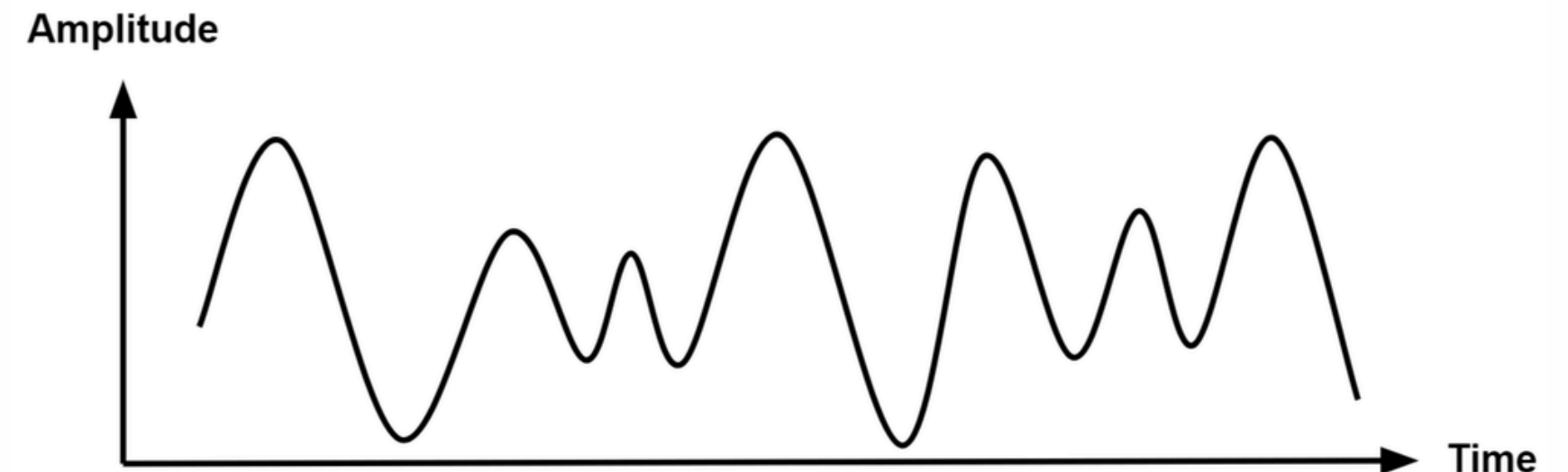
Digital Transmission

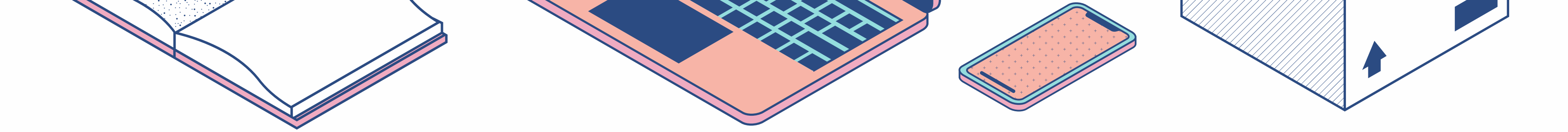
- process of sending information in a **discrete** or **binary form**, using bits (composing 0s and 1s)
- **Advantages:** High Accuracy, Efficiency, Multiplexing, Compression, Security



Analog Transmission

- sending information in a continuous wave form, analogous to the original data
- **Advantages:** Simplicity, Signal Fidelity, Real-Time Processing



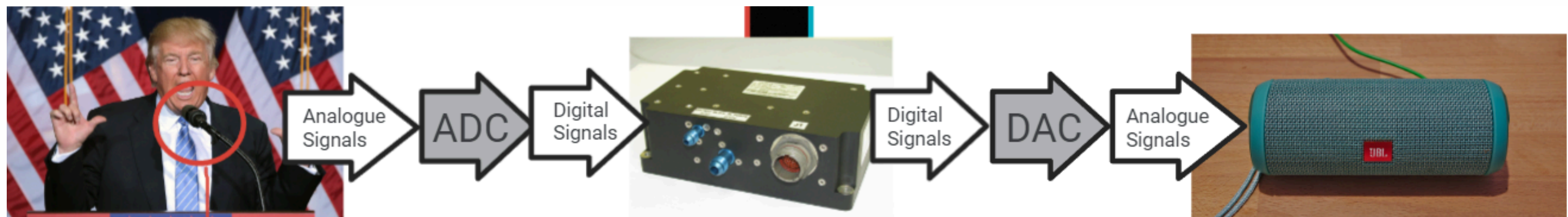


Digital-to-Analog Converter

- converts a digital input signal into an analog output signal
- translates digital values into continuous voltage levels to recreate the original analog signal

Analog-to-Digital Converter

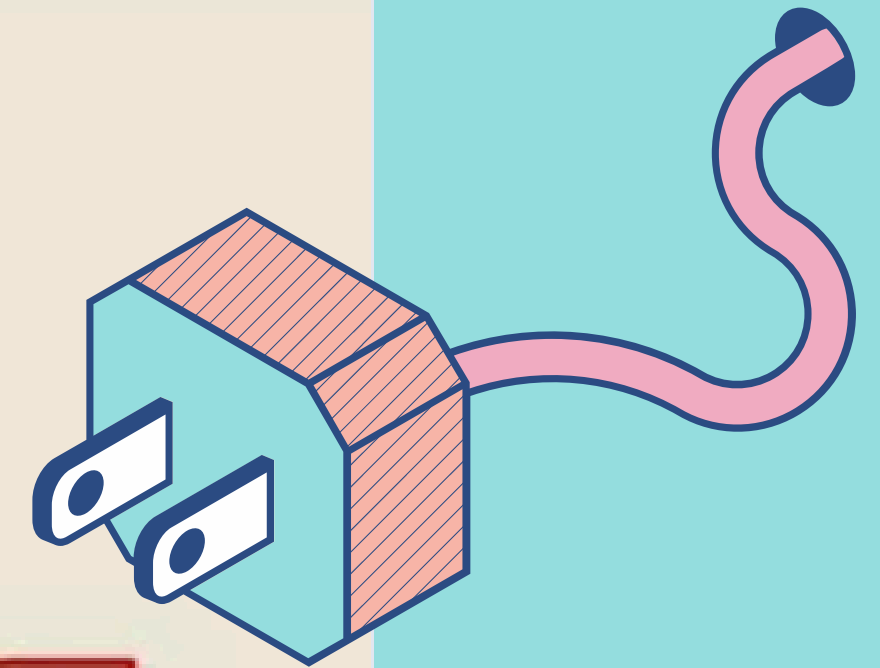
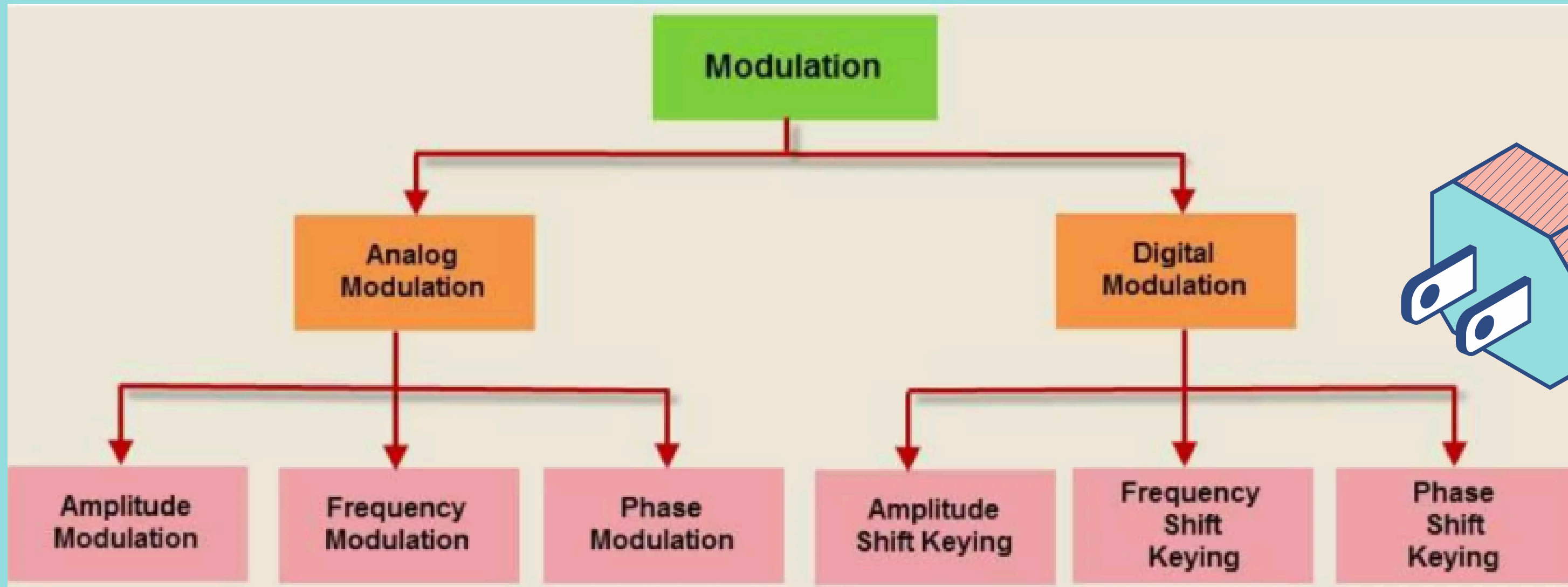
- allows digital circuits to interface with the real world by encoding an analogue signal into a binary code
- allows analog signals from sensors or audio inputs to be processed by digital systems (computers or microcontrollers)

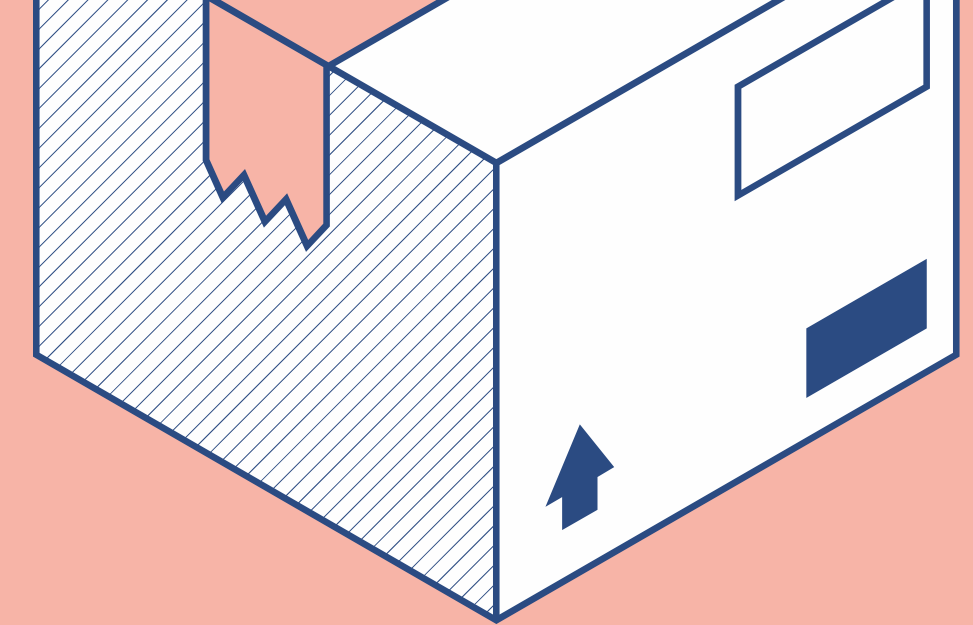




Modulation

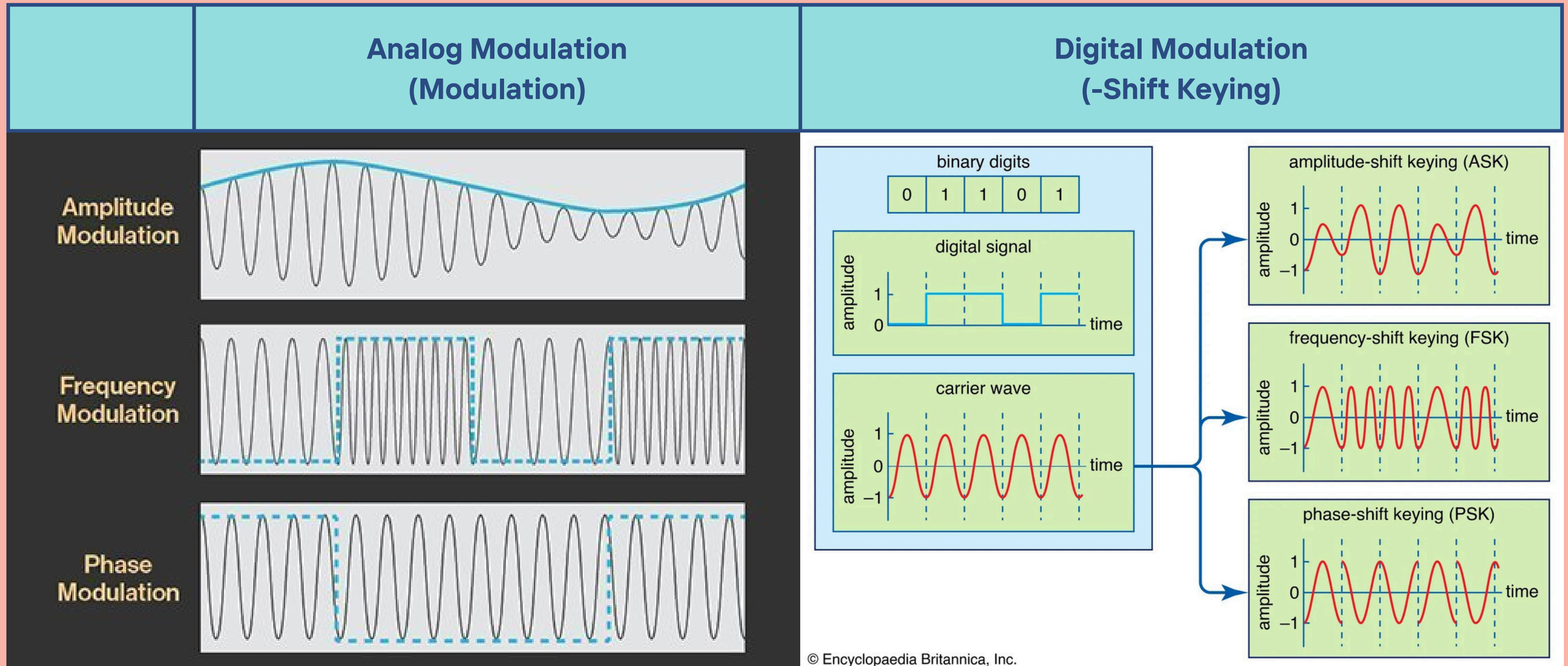
Modulation is the technique of adding information to an optical or electrical carrier signal in order to transform data into radio waves. A carrier signal is one that has a consistent frequency and height, or amplitude, on its waveform.





Modulation Techniques

	Analog Modulation (Modulation)	Digital Modulation (-Shift Keying)
Amplitude	amplitude of the carrier signal is varied in proportion to the analog message signal	amplitude of the carrier signal is varied in accordance with the digital data
Frequency	frequency of the carrier signal is varied in proportion to the analog message signal	frequency of the carrier signal is varied to represent digital data.
Phase	phase of the carrier signal is varied in proportion to the analog message signal	phase of the carrier signal is shifted to represent digital data





ATENEO DE ZAMBOANGA UNIVERSITY
CIT.016 | DATA COMMUNICATION

THANK YOU

FOR LISTENING

